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archeryutils: a Python package for archery calculations

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Software

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Summary

The sport of archery lends itself well to mathematical analysis and indeed there are several approaches to modelling performance across different distances and target types.

archeryutils is a Python package that provides base types and functions for developing code with archery applications, as well as implementations of popular mathematical models used in the sport.

It is used in multiple applications including the development of websites and apps for archers, conducting performance analysis for elite archers and governing bodies, and in ongoing sports science research.

Statement of Need

In the sport of archery athletes shoot targets at a variety of distances. Further, these targets can be of different sizes and have different scoring systems associated with them. The overall goal is always the same, however; to shoot arrows as accurately and precisely as possible. For the purposes of analysis, it is therefore useful to be able to extract some comparative measure of performance across all targets and distances. As a precision-based sport, archery lends itself well to mathematical modelling with examples in literature being Lane (2013), Park (2014), and the 2023 handicap system by Archery GB (Atkinson & Archery GB, 2023). Although these models are well described in their respective papers, no software implementations have been made publicly available, presenting barriers to further research and development and reproducibility. *archeryutils* has been developed to remove these barriers.

Archery score data is increasingly readily available in electronic formats, be it from competition results (e.g. on ianseo, tamlynscore, etc.) or the various scoring applications used by archers of all levels. The *archeryutils* API is easily combined with Python data-processing packages, for example pandas (McKinney & others, 2011; The pandas development team, 2024) or polars (Ritchie Vink et al., 2025), enabling users to perform analysis on these datasets without needing to reimplement models themselves.

Software Description

archeryutils is a Python package distributed through the Python Package Index (PyPI) with full documentation, including the API, provided online at archeryutils.readthedocs.io

At the core of *archeryutils* are classes representing the core components of the sport. Perhaps the most notable of these are the `Target`, providing a representation of a target of a particular size, distance, and face type, and the `Round`, which builds upon this to represent of a series of targets shot at with a number of arrows. As well as a constructors for creating generic

targets and rounds, *archeryutils* comes with many popular national and international rounds predefined for users. These components provide a starting point for a variety of applications.

The implementation of the mathematical models for precision can be found in the `handicaps` module. Available models include both Archery GB handicap schemes of Lane (2013) and the updated 2023 model and both Archery Australia models of Park (2014), all streamlined into a common API. There is also the option for users to define custom models by building on the abstract base class provided.

One other notable component of *archeryutils* is the `classifications` module. This provides an implementation of the Archery GB classification progression schemes in which archers are categorised and rewarded for their performance depending on their age, gender, and bowstyle/division across the target indoor, target outdoor, and field disciplines. *archeryutils* was used to develop these schemes and generate the associated data, so can be regarded as the definitive source.

Theory - Precision Models

Whilst specific implementations vary, models for an archer's precision (also called 'handicap' or 'skill level' in literature) take following form:

$$\sigma_\theta(h, d) = \theta(h) \cdot g(d)$$

where h is a number quantifying the archer's handicap or skill and $\theta(h)$ is a function that relates h to the angular deviation of their shots. This can then be mapped downrange to the target at distance d, with $g(d)$ quantifying the effect of 'excess dispersion'.[1]

The angular deviation of arrows can be converted into a radial distribution of on the target face by multiplying by the distance to the target:

$$\sigma_r(h, d) = \sigma_\theta(h, d) \cdot d$$

This defines a normal distribution for probability density of where an archer's arrows will land on the target which can then be integrated with a function for scoring to get an expected score.

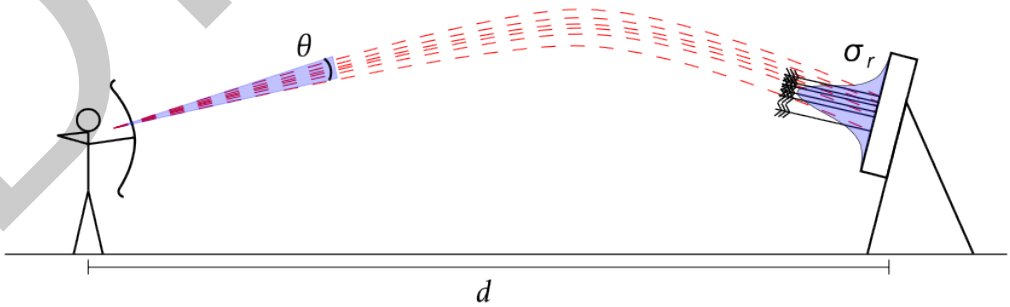

Figure 1: Schematic showing key components of models.

These models make key assumptions that the archer is always accurate (dispersed about the centre of the target) and that the archer's arrows are normally distributed with a single value of σ_θ (no systematic errors).

[1] How angular deviation grows in excess of a linear relation to distance

The earliest such model is discussed in a series of articles (Lane, 1979a, 1979b) with equations made available in Lane (2013). It was used to construct the original handicapping scheme of Archery GB, the national governing body of archery in the United Kingdom. An alternative model proposed by Park (2014) is used by Archery Australia, and forms the basis of their national ranking system. It has also been applied in studies of competition performance and equipment (Park, 2021, 2023).

Examples of Use

archeryutils is used in a range of applications. Its origins lie in the development of the 2023 handicap model for Archery GB where it was used in conjunction with competition data to develop and tune model parameters (Atkinson & Archery GB, 2023). It was subsequently used in the development of new Archery GB classification systems (Atkinson et al., 2023, 2024; Atkinson & Archery GB, 2025). The software provides the backend for the community website archerycalculator, as well as being used in several other applications and websites.

The models implemented by *archeryutils* have previously been used in comparative analyses of athletes' performance and equipment, and there is ongoing research in this area that uses the software.

Acknowledgments

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